shelves can help. Shure's recording guidance emphasises that room acoustics and microphone placement strongly affect the balance of direct and reflected sound captured by the microphone.

Use this room-preparation sequence:

- **Step 1:** Choose the smallest comfortable furnished room available.
- **Step 2:** Close windows and doors and silence devices that are not essential.
- **Step 3:** Move away from bare walls and large glass surfaces.
- **Step 4:** Put soft material behind and beside the speaker if the room sounds lively.
- **Step 5:** Place noisy equipment, including a desktop PC, behind the microphone rather than directly in front of it when the microphone's pickup pattern allows.
- **Step 6:** Record 20 seconds and listen through headphones before setting up the entire session.

If switching off a fan or air-conditioner makes the room unbearable, cool it beforehand, use the quietest setting available, or record in shorter blocks.

Get close to the microphone

Distance is one of the cheapest audio upgrades available.

For close spoken-word recording, many broadcast-style microphones are designed to work only a few centimetres from the mouth. Shure recommends speaking about 1–6 inches, or roughly 2.5–15cm, from models such as the SM7B and MV7 to maximise direct voice and reject more off-axis sound.

Do not blindly copy that distance for every microphone; check its manual. But the principle is important: the closer the microphone is to the voice, the less gain you generally need and the less prominent the room becomes.

Position the microphone slightly to one side of the mouth rather than directly in the path of your breath. This reduces harsh bursts on "p" and "b" sounds. A pop filter or foam windscreen adds another layer of protection.

Once you find a position that sounds good, keep it consistent. Shure specifically recommends maintaining a consistent microphone position to help maintain consistent levels.

Set gain without chasing loudness

Gain determines how strongly the microphone signal is amplified before it is recorded. Too low and you may amplify hiss later. Too high and loud syllables can clip.

For conventional 16-bit or 24-bit recording, leave headroom. RØDE's current gain-staging guidance recommends starting with the preamp gain low, increasing it while monitoring, and aiming for peaks between approximately -12dB and -6dB. It also warns that distortion introduced during recording cannot be fully repaired afterwards. Audacity's manual similarly advises making a test recording and aiming for peaks around -6dB while checking that the waveform is not clipping.

BUILD A BACKUP HABIT

A loose cable, full memory card, dead battery or software crash can quietly ruin an hour.

Create redundancy before every serious session. Portable recorders increasingly support simultaneous or secondary recording options; for example, Zoom's H5studio can function as a USB audio interface while simultaneously recording a backup to microSD.

Your backup does not need to be identical to the primary recording. A phone voice recorder placed safely near the speakers is much better than having no second copy.

Use a pre-recording checklist: storage available, batteries charged, power connected, cables secure, correct microphone selected, tracks armed, headphones working and backup recording running.

After recording, copy the files to a computer immediately. Keep the originals untouched and create a second copy on another drive or cloud storage before editing.

Set levels like this:

- **Step 1:** Put on headphones and position the microphone correctly.
- **Step 2:** Ask the speaker to deliver an energetic sentence at the loudest level they are likely to use.
- **Step 3:** Begin with gain low and increase it gradually.
- **Step 4:** Watch the meter during louder words. Keep peaks safely below 0dB.
- **Step 5:** Record 20–30 seconds, including normal speech, laughter and an emphatic sentence.
- **Step 6:** Play it back before starting the episode.

Never set gain while the guest murmurs "testing, one, two" and then assume it will survive their excited storytelling voice.

What about 32-bit float?

Some newer portable recorders and interfaces support 32-bit float recording. Its very large dynamic range can make unexpectedly loud or quiet digital recordings far more recoverable in post-production. Zoom explains that 32-bit float files allow large level changes to be corrected later, whereas clipping in conventional lower-bit-depth files can be irreversible.

That makes 32-bit float useful for unpredictable field recordings or guests whose volume changes dramatically. It does not make microphone technique irrelevant: a distant microphone still captures too much room, wind can still overwhelm recordings, and lavaliers can still rub against clothing.